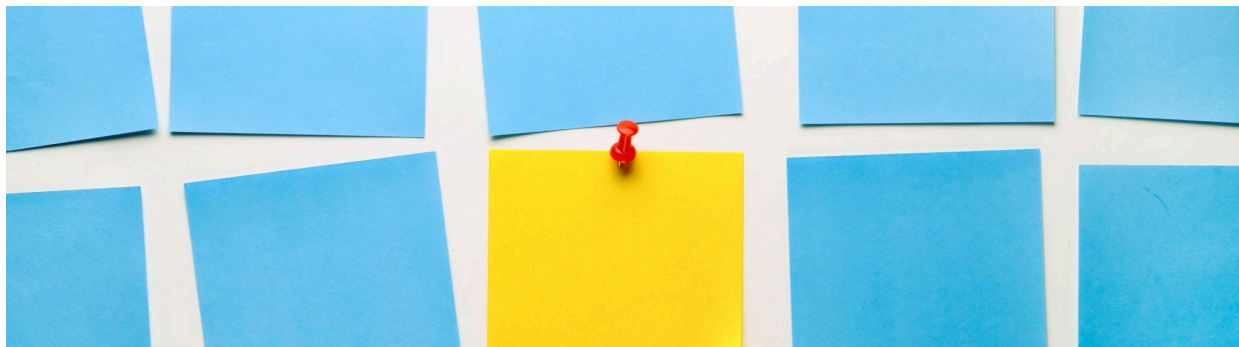




NFL 4th Down Success Probability – Executive Summary

Status: Completed

Timing: Mar 4, 2026 – Apr 15, 2026

Owner: Willber Escalante

Objective

Analyze NFL play-by-play data (2009–2018) to build a model that estimates the probability that a 4th-down punt, field goal, or go-for-it attempt will succeed given game context, and use it to evaluate how conservative or aggressive coaches are in key situations.

Data and approach

- Data: Kaggle NFL Play-by-Play dataset, 2009–2018.
- Target: 3 decision classes (punt, field goal, go for it); success defined as scoring or gaining a new set of downs.
- Features: engineered game-context buckets such as field position (yardline_100, red zone), distance to go, time and score situation, and late-game/OT flags.
- Models: three XGBoost classifiers (baseline with leakage, class-balanced, and tuned final model) evaluated with accuracy and macro F1 to balance all three decision types.

Key findings (behavior)

- Coaches choose the conservative option most often: punts on ~60.8% of 4th downs, field goals on 22.8%, and “go for it” on only 16.4%.
- Only 25.8% of 4th-down plays result in a successful outcome (points or a first down).
- Field goals succeed about 85% of the time, while go-for-it attempts succeed about 36.2% of the time.

Typical patterns:

- Field goals are concentrated in the red zone with longer distances to go.
- Punts mostly occur in the offense's own territory, across many distances.
- Go-for-it attempts cluster in opponent territory, usually with ≤ 1 yard to go and more often earlier than later in the game.

Key findings (model)

- Model 1 (leakage-prone baseline): ~90% accuracy, macro F1 $\approx 84\%$, but uses ID-like features and overestimates performance, especially on go-for-it plays.
- Model 2 (balanced, no IDs): ~88% accuracy, macro F1 $\approx 83.4\%$; removing leakage reduces headline scores but improves recall on go-for-it decisions to ~68% at the cost of some precision.
- Model 3 (final tuned model): ~88.6% accuracy, macro F1 $\approx 83.6\%$; maintains ~67% precision and slightly higher recall (~68.3%) for go-for-it plays while using only realistic, pre-snap context features.
- Most important features across models are field position (red zone vs non-red-zone, yardline_100, and being in opponent territory with short yardage), confirming that location and distance dominate 4th-down decision quality.

Recommendation

Adopt Model 3, the tuned XGBoost model trained on unleaked, pre-snap context features with inverse-frequency class weights, as the final model for this project. It:

- Uses only information available in real time.
- Trades a small amount of raw accuracy for better macro F1 and recall on the rare but critical “go for it” decisions.
- Treats punt, field goal, and go-for-it as equally important, making it suitable for evaluating coaching aggressiveness and for powering a future sideline decision tool.